Recall we defined $\mathbb{Q}$ as $\mathbb{Q} := (\mathbb{Q}, +, \cdot, <)$.

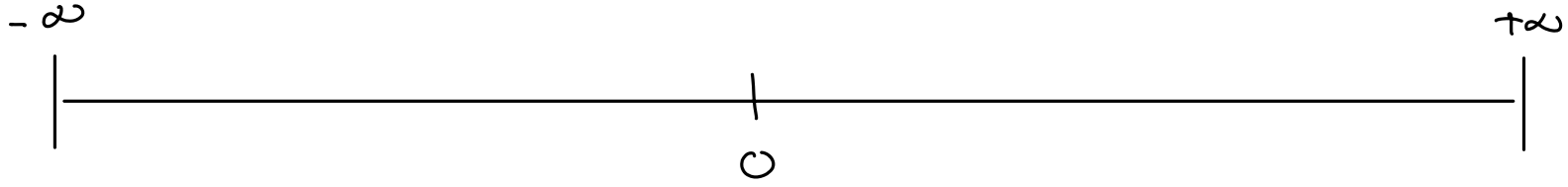
2.1 Who ordered these?

Unlike $\mathbb{N}$ and $\mathbb{Z}$, the rational numbers have the special property that between any two distinct $x, y \in \mathbb{Q}$, there exists $z \in \mathbb{Q}$ such that $x < z < y$. This is clear, and z can be explitly constructed via

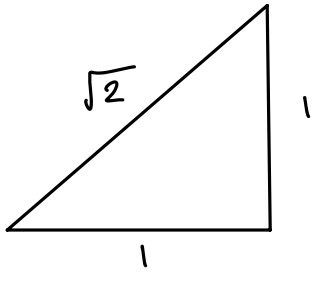
$$z = \frac{x+y}{2}.$$

Given this property, it would seem reasonable that we can represent $\mathbb{Q}$ by an infinite numberline with no gaps.

$\mathbb{Q}$.



Consider the following



Theorem 2.4: The $\sqrt{2}$ is not rational.

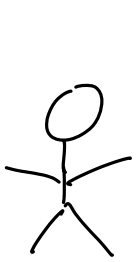
Proof: Suppose towards a contradiction that $0 < \sqrt{2}$ is rational. Hence, there exists $a, b \in \mathbb{N}$ such that $\sqrt{2} = a/b$. Assume, without the loss of generality (WLOG) that a/b is irreducible. That is, there are no common divisors of a and b other than 1. Then

$$2 = a^2/b^2 \Rightarrow 2b^2 = a^2.$$

Hence, a^2 is even. Thus a is even. Then $a = 2k$ for some $k \in \mathbb{N}$. Thus $a^2 = 4k^2 \Rightarrow 2b^2 = 4k^2$. That is, $b^2 = 2k^2$. So, b^2 is even and then b is even. Hence, a/b is reducible. Contradiction. ∎

Either we take $\sqrt{2}$ as non-existent and only use geometry, or we try create a number system beyond measurement in which $\sqrt{2}$ exists.

Consider  $(s = \frac{1}{2}gt^2)$ $g = 10ms^{-2}$.



$$10 = \frac{1}{2}(10)t^2 \Rightarrow t = \sqrt{2}.$$

Defn 2.5: (Ordered Field)

An ordered field $\mathbb{F}$ is a non-empty set with two binary operations, addition $(+)$ and multiplication $(\cdot)$, as well as two relations $(<)$ and $(=)$, such that

i) (Commutative Law) If $a, b \in \mathbb{F}$ then $a+b = b+a$ and $a \cdot b = b \cdot a$.

ii) (Distributive Law) If $a, b, c \in \mathbb{F}$ then $a \cdot (b+c) = a \cdot b + a \cdot c$.

iii) (Associative Law) If $a, b, c \in \mathbb{F}$, then $a+(b+c) = (a+b)+c$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

iv) (Identity Law) There exists elements $0, 1 \in \mathbb{F}$ such that $a + 0 = a$ and $a \cdot 1 = a$ for all $a \in \mathbb{F}$.

v) (Inverse Law) For every $a \in \mathbb{F}$, there exists $-a \in \mathbb{F}$ such that $a + (-a) = 0$. If $a \neq 0$, there exists $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1$.

vi) (Order Axioms) Take $a, b \in \mathbb{F}$

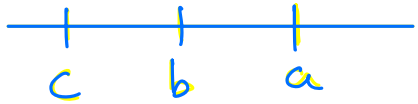
(a) Either $a < b$, $b < a$ or $a = b$.

(b) If $a < b$ then for all $c \in \mathbb{F}$ it follows that $a + c < b + c$.

(c) If $0 < a$ and $0 < b$ then $0 < a \cdot b$.

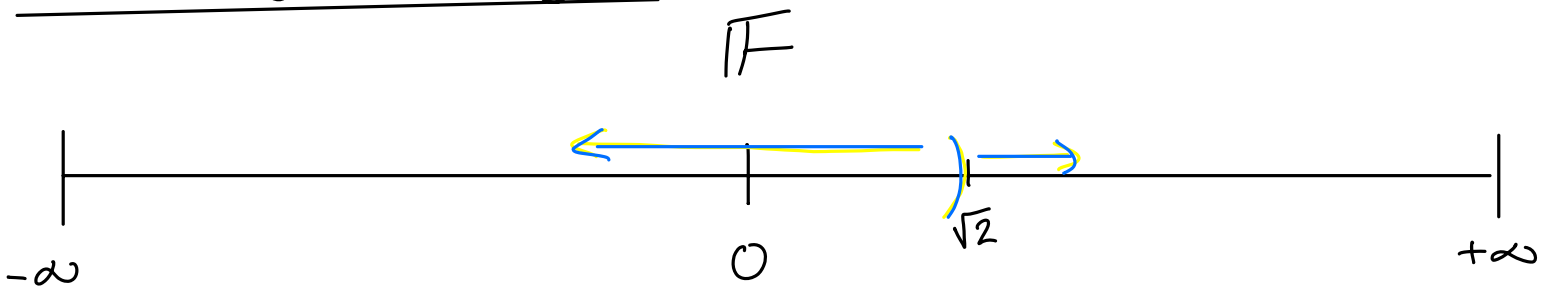
(d) If $b < a$ and $c < b$, with $c \in \mathbb{F}$, then $c < a$.

Pic:



A <u>field</u> is simply an ordered field with the order axioms removed.

2.2 Filling in the gaps



(2.6)

<u>Defn</u>: (Bounded Set)

Let S be a non-empty subset of an ordered field $\mathbb{F}$. Then S is said to be <u>bounded above</u> if there

exists an $\alpha \in \mathbb{F}$ such that $x \leq \alpha$ for all $x \in S$.

We call α an <u>upper bound</u> for S.